

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250269991

Kind Code

A1

Publication Date

August 28, 2025

Inventor(s)

Gerdes; John W. et al.

ROTOR HUB WITH OFFSET TEETER AXIS

Abstract

A teetering rotor hub is described. The teetering rotor hub may be able to couple to multiple rotor blades and to a rotor. The teetering rotor hub may be associated with an aerial system. The teetering rotor hub may be able to rotate about a rotor spin axis of an associated motor. The teetering rotor hub may couple to the rotor or shaft. In some embodiments, the rotor hub may couple to a pair of rotor blades. The teetering rotor hub may be able to teeter along a second axis that is offset from the rotor spin axis, where the offset may be between zero and ninety degrees. By varying a speed signal provided to the rotor, the rotor blades coupled to the rotor hub may be moved in various ways to generate flapping motions that are able to control the direction of movement of the aerial system.

Inventors: Gerdes; John W. (Bel Air, MD), Kang; Hao (Abingdon, MD), Hrynuk; John T. (Joppa, MD), Rigas; Elias J. (Newark, DE)

Applicant: U.S. Army DEVCOM Army Research Laboratory (Adelphi, MD)

Family ID: 1000007797147

Appl. No.: 18/587575

Filed: February 26, 2024

Publication Classification

Int. Cl.: B64U30/297 (20230101)

U.S. Cl.:

CPC B64U30/297 (20230101); B64U2201/00 (20230101)

Background/Summary

FIELD OF THE INVENTION

[0002] The invention relates to control of aerial systems.

BACKGROUND OF THE INVENTION

[0003] Existing aerial systems may utilize complicated and expensive equipment such as a swashplate to achieve directional control of the aerial system via a rotor.

[0004] Thus, there is a need for ways to achieve directional control via a simplified rotor interface.

BRIEF SUMMARY OF THE INVENTION

[0005] Some embodiments may provide a teetering rotor hub. The rotor hub may be able to couple to multiple rotor blades and to a rotor. The rotor hub may be associated with an aerial system.

[0006] The rotor hub may be able to rotate about a first axis that is parallel to a rotor spin axis of an associated motor. The rotor hub may couple to the rotor or shaft. In some embodiments, the rotor hub may couple to a pair of rotor blades.

[0007] The rotor hub may be able to teeter along a second axis that is offset from the first axis, where the offset may be between zero and ninety degrees.

[0008] By varying a speed signal provided to the rotor, the rotor blades coupled to the rotor hub may be moved in various ways to generate flapping motions that are able to control the direction of movement of the aerial system.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0009] The novel features of the disclosure are set forth in the appended claims. However, for purpose of explanation, several embodiments are illustrated in the following drawings.

[0010] FIG. 1 illustrates a top plan view of a rotor hub of one or more embodiments described herein;

[0011] FIG. 2 illustrates a front elevation view of a rotor hub, in which the teeter axis is not offset relative to the z-axis;

[0012] FIG. 3 illustrates a front elevation view of a rotor hub of one or more embodiments described herein, in which the teeter axis is offset from the z-axis;

[0013] FIG. 4 illustrates a right-side elevation view of a rotor hub of one or more embodiments described herein, in which the rotor hub teeters along an offset teeter axis;

[0014] FIG. 5 illustrates a front elevation view of a rotor hub system of one or more embodiments described herein, in which the rotor hub is coupled to a motor;

[0015] FIG. 6 illustrates a right-side elevation view of a rotor hub system of one or more embodiments described herein, in which the rotor hub teeters about the offset teeter axis;

[0016] FIG. 7 illustrates an example overview of one or more embodiments described herein, in which a constant rotor speed is applied;

[0017] FIG. 8 illustrates an example overview of one or more embodiments described herein, in which a variable rotor speed associated with a first movement direction is applied;

[0018] FIG. 9 illustrates an example overview of one or more embodiments described herein, in which a variable rotor speed associated with a second movement direction is applied;

[0019] FIG. 10 illustrates a flow chart of an exemplary process that implements flight control via the rotor hub of one or more embodiments described herein; and

[0020] FIG. 11 illustrates a schematic block diagram of one or more exemplary devices used to implement various embodiments.

DETAILED DESCRIPTION OF THE INVENTION

[0021] The following detailed description describes currently contemplated modes of carrying out exemplary embodiments. The description is not to be taken in a limiting sense, but is made merely for the purpose of illustrating the general principles of some embodiments, as the scope of the

disclosure is best defined by the appended claims.

[0022] Various features are described below that can each be used independently of one another or in combination with other features. Broadly, some embodiments generally provide a rotor hub with an offset teeter axis. The offset teeter axis allows for operation without the use of a swashplate.

[0023] FIG. 1 illustrates a top plan view of a rotor hub **100** of one or more embodiments described herein. As shown, the rotor hub **100** may include a rotor hub hinge **110**, a teeter axle **120**, and a rotor coupling **130**. In this example, the rotor hub **100** is coupled to a pair of rotor blades **140**, but the rotor hub **100** may be coupled to various other sets of blades, propellers, and/or other appropriate components in any appropriate configuration and/or quantity.

[0024] Rotor hub **100** may be a mechanical device or component that is able to couple to a rotor (e.g., a cylindrical rotating member of a motor) via rotor coupling **130**. Rotor hub **110** may be able to couple to rotor blades **140** and/or other appropriate components. Rotor hub **100** may include various appropriate components and/or elements in addition to, and/or in place of, those shown. For example, rotor hub **100** may include a housing or other enclosure. Rotor hub **100** may include various couplings or connectors that may allow the rotor hub **100** to interface with various rotors, rotor blades **140**, and/or other appropriate components.

[0025] Rotor coupling **130** may include a shaft, axle, cylindrical cavity, and/or other appropriate components that may be coupled to a rotor. Rotor coupling **130** may include appropriate clamps, fasteners, adhesives, etc. as may be used to couple the rotor hub **100** to the rotor shaft.

[0026] Each rotor blade **140** may be any type of blade that is appropriate for the aerial system associated with the rotor hub **100**. Such blades may be various sizes and/or shapes, include various different materials, have different interfaces or connectors, and/or include various features or attributes. In some cases, the rotor blades **140** may be fixedly coupled to the rotor hub hinge **110** such that the rotor blades **140** do not move relative to the rotor hub hinge **110**.

[0027] X-axis **150** is parallel to the teeter axle **120** in this example. Y-axis **160** is perpendicular to x-axis **150**. In this example, x-axis **150** and y-axis **160** may be perpendicular to a shaft or axle associated with rotor coupling **130** (and the associated rotor shaft).

[0028] FIG. 2 illustrates a front elevation view of a rotor hub **100**, in which the teeter axis **210** is not offset relative to the z-axis **220**. The tip-path-plane of rotor blades **140** (i.e., the tip-path plane of the rotor blades **140** is parallel to teeter axis **210** and x-axis **150**). The teeter axis **210** (or “lag-coupled” teeter axis) may run along a center or rotational axis of teeter axle **120**. The tip-path-plane of the rotor blades **140** is perpendicular to a rotational axis of the motor rotating shaft **230** (included for clarity) in this non-offset configuration. Z-axis **220** may be perpendicular to x-axis **150** and y-axis **160**. Z-axis **220** may be parallel to the rotational axis of the motor rotating shaft **230**. If the teeter axis **210** is parallel to x-axis **150** and perpendicular to z-axis **220** as shown, this configuration is a typical teetering rotor where the hinge allows the blades to flap together in opposite directions like a seesaw.

[0029] FIG. 3 illustrates a front elevation view of a rotor hub **100** of one or more embodiments described herein, in which the teeter axis **310** is offset relative to the z-axis **220**. In this configuration, the tip-path-plane of the rotor blades **140** may remain parallel to the x-axis **150** and y-axis **160** (and non-offset teeter axis **210**), while the offset teeter axis **310** may be offset from the non-offset teeter axis **210** (and x-axis **150**) by some offset angle (θ) **320**.

[0030] Offset, or lag-coupled, teeter axis **310** has a forty-five-degree offset angle (θ) **320** relative to non-offset teeter axis **210** (and x-axis **150** and z-axis **220**) in this example. Offset angle (θ) **320** may be greater than zero degrees and less than ninety degrees.

[0031] Rotor hub hinge **110** may be a teetering hinge that is able to teeter about a teeter axle **120** or other rotating element that is offset from a plane defined by the path of rotors **140**. Teeter axle **120** may be aligned with offset teeter axis **310** (or any other offset teeter axis having between zero and ninety degrees of offset), as shown in this example. Rotor hub hinge **110** may include various sets of sub-elements, such as axles or shafts (e.g., teeter axle **120**), guides, stops, cavities or receptacles

(e.g., a cylindrical cavity that receives a portion of a rotor), and/or other such elements that may provide rotational movement about one or more axes. Rotor hub hinge **110** may couple to a rotor via rotor coupling **130** such that the rotor hub hinge **110** may teeter across an offset axis (e.g., offset teeter axis **310**) while rotating about the rotor axis of rotation.

[0032] Teeter axle **120** may include an axle, shaft, hinge, or similar element that allows the rotor hub hinge **110** to rotate about a teeter axis (e.g., offset teeter axis **310**). Teeter axle **120** may include, utilize, and/or otherwise be associated with various elements, such as receptacles or cavities, guides, stops, etc., that may define the range of motion of the teeter axle **120** and the axis of rotation (e.g., offset teeter axis **310**) relative to other components of the rotor hub **100**.

[0033] Because of the incline of the lag-coupled teeter axis **310** out of the plane including x-axis **150** and y-axis **160** (and relative to z-axis **220**), the teetering motion is augmented through a kinematic coupling between teetering (flapping) and azimuthal lag (where the “azimuthal” axis is parallel to z-axis **220** and may run along the rotational axis of motor rotating shaft **230**). Due to the finite inertia of the rotor blades **140**, any acceleration along the azimuthal axis must introduce a lag in the same direction, which will then couple into flapping motion as a function of the offset angle (θ) **320**.

[0034] Practically, the result of the coupling described above is that the tip-plane-path of the rotor blades **140** may be pointed away from the z-axis **220** by varying the timing of azimuthal inertial loads (e.g., by varying the speed of rotation of rotor **220** throughout each rotation). By superimposing a sinusoidal velocity at the rotor frequency with a steady rotational velocity, a cyclic oscillation in blade flap may be achieved due to the sinusoidal lag motion of the rotor blades **140**. The cyclic flap of the rotor blades **140** tilts the rotor blade tip-path-plane in some new direction offset from the x-axis **150** and/or y-axis **160**, thus generating a component of the rotor thrust in the rotor hub plane including x-axis **150** and y-axis **160** and also moments on the rotor hub **100**. These modified hub loads may be used to achieve altitude control for an aerial system (and/or other associated system, device, or set of components). By changing the magnitude of the superimposed sinusoidal velocity, the magnitude of the loads in the rotor hub plane can be adjusted. The direction of the hub forces and moments may be controlled by changing the phase of the sinusoidal velocity relative to the rotor hub **100** frame.

[0035] FIG. 4 illustrates a right-side elevation view of a rotor hub **100** of one or more embodiments described herein, in which the rotor hub **100** teeters about the teeter axle **120** along offset teeter axis **310** (e.g., a teeter axis having a forty-five-degree offset angle (θ) **320** running into the page). Z-axis **220** may be parallel to the rotational axis of the motor rotating shaft **230**.

[0036] As shown, the rotor blades **140** may follow various tip-path-planes, in this example ranging from negative thirty to positive thirty degrees about offset teeter axis **310**. Different embodiments may have different teetering ranges than shown (e.g., plus-minus forty-five degrees, fifteen degrees, etc.).

[0037] FIG. 5 illustrates a front elevation view of a rotor hub system **500** of one or more embodiments described herein, in which the rotor hub **100** is coupled to a motor **510**. In this example, rotor hub **100** may include a rotor hub hinge **110** and a rotor coupling **130**, as well as any or all other components described above.

[0038] Rotor hub system **500** may include rotor hub **100** and a motor **510**. Rotor hub system **500** may be associated with an aerial system such as an unmanned aerial vehicle (UAV) or unmanned aerial system (UAS). The rotor hub system **500** may be coupled to a housing or exterior surface of the UAV or UAS (e.g., the top surface of a UAV), where such a surface may be parallel to the x-axis **150** and y-axis **160** in this example. Rotor hub system **500** may include various power couplings and/or communicative couplings that may allow the rotor hub system **500** to interact with an associated resource such as a UAV (e.g., by receiving power from the UAV, sending sensed or measured information to the UAV, etc.).

[0039] Rotor coupling **130** in this example may include a frame that couples to motor **510**. The

frame may have a modified “L” shape, with a horizontal member coupled to motor **510** and a vertical member that extends out from the horizontal member at a right angle as shown. Rotor coupling **130** may include a hinge coupling shaft **520** that is aligned with offset teeter axis **310**. In this example, offset teeter axis **310** has an offset angle (θ) **320** of forty-five degrees relative to a non-offset teeter axis **210**, where the offset angle (θ) **320** may be greater than zero degrees and less than ninety degrees. In this example, the rotor blades **140** may be mounted at a matching angle relative to the rotor hub hinge **110** (negative forty-five degrees in this example), as shown, such that the rotor blades **140** may follow a tip-path-plane parallel to non-offset teeter axis **210** when the rotor hub hinge **110** is angled as shown. Rotor hub hinge **110** may include a cylindrical cavity that is able to accept at least a portion of hinge coupling shaft **520** such that rotor hub hinge **110** is able to rotate about the hinge coupling shaft **520** (or offset teeter axis **310** as described above in reference to FIG. 4). Returning to the example of FIG. 5, the hinge coupling shaft **520** and cylindrical cavity may be components of the teeter axle **120**.

[0040] Motor **510** may be a brushless motor that is able to change speeds very quickly. For example, speed changes may be synchronized with a rotation point relative to the rotor of motor **510** (e.g., rotation points differing by one-degree, one-half degree, etc.) and the rotor coupling **130** and associated rotor hub **100**. Motor **510** may include a stationary motor portion **530** and a rotating motor portion **540**. The stationary motor portion **530** may be fixedly coupled to an associated resource such as a UAV, test fixture, other non-moving substrate, etc. Rotating motor portion **540** may be able to rotate relative to stationary motor portion **530** about rotor axis **550**. Rotor axis **550** may be parallel to z-axis **210** and may be aligned with a rotor of the motor **510** (e.g., motor rotating shaft **230**), or a center point of a rotating portion of the motor **510** such as rotating motor portion **540**. In this example, motor **510** may have a generally cylindrical shape and rotor axis **550** may be aligned with a center axis of the cylindrical shape. When powered, all components except for the stationary motor portion **530** rotate together about the vertically oriented rotor axis **550** (or “azimuthal” axis) of the motor.

[0041] FIG. 6 illustrates a right-side elevation view of a rotor hub system **500** of one or more embodiments described herein, in which the rotor hub teeters about offset teeter axis **310**, which is aligned with hinge coupling shaft **520** in this example. In this example, rotor hub hinge **110** is able to teeter plus/minus thirty degrees about the offset teeter axis **310**.

[0042] FIG. 7 illustrates an example overview of one or more embodiments described herein, in which a constant rotor speed is applied. When rotational velocity is constant, there is no lag force and centrifugal force keeps the tip-path-plane of the rotor blades **140** level, such as when a UAV hovers (or ascends or descends along a vertical axis).

[0043] Front elevation view **710** shows the tip-path-plane **720** of the rotor blades **140** as being perpendicular to rotor axis **550**. Similarly, left-side elevation view **730** shows the tip-path-plane **720** of the rotor blades **140** as being perpendicular to rotor axis **550**. Speed signal **740** shows a constant output level, where the speed signal **730** may be associated with a voltage or control signal applied to a motor (e.g., motor **510**) and is represented over a full rotation about the rotor axis **550**.

[0044] FIG. 8 illustrates an example overview of one or more embodiments described herein, in which a variable rotor speed associated with a first movement direction is applied. This example may be associated with forward movement (that may be coupled with increasing, decreasing, or constant altitude).

[0045] Front elevation view **810** shows the tip-path-plane **820** of the rotor blades **140** as flapping downward toward the front of an associated UAV or other platform and upward toward the rear of the associated UAV or other platform, such that forward motion is achieved. Left-side elevation view **830** shows the tip-path-plane **820** of the rotor blades **140** as being tipped forward relative to the UAV or other platform, resulting in forward speed (along the zero-degree azimuth angle position), Speed signal **840** includes the constant speed signal **740** and a superimposed sinusoid aligned with a one-hundred-eighty-degree rotational point (and zero-degree direction of movement

relative to the azimuth angle and the UAV or other platform). The rotational point may be defined relative to a position of the rotor blades **140** during rotation about the rotor axis **550**.

[0046] As shown, there is a clear offset between the rotor axis **550** and the tip-path-plane of the rotor blades **140** (fifteen degrees in this example). Thus, it is feasible to achieve directional control of a UAV or similar platform.

[0047] The flap deflection that arises from lag in the azimuthal direction is shown clearly. It is notable that the exact coupling angle between lag and flap is an adjustable feature of this design that arises from the magnitude of the angle between the lag-coupled teetering axis and the teeter axis.

[0048] FIG. **9** illustrates an example overview of one or more embodiments described herein, in which a variable rotor speed associated with a second movement direction is applied. This example may be associated with rightward movement (that may be coupled with increasing, decreasing, or constant altitude),

[0049] Front elevation view **910** shows the tip-path-plane **920** of the rotor blades **140** as being tipped leftward relative to the UAV or other platform, resulting in rightward speed (along the two-hundred-seventy-degree azimuth angle position). Left-side elevation view **930** shows the tip-path-plane **920** of the rotor blades **140** as flapping downward toward the left side of an associated UAV or other platform and upward toward the right side of the associated UAV or other platform, such that rightward motion is achieved. Speed signal **940** includes the constant speed signal **740** and a superimposed sinusoid aligned with a ninety-degree rotational point (and two-hundred-seventy-degree direction of movement relative to the azimuth angle and the UAV or other platform). The rotational point may be defined relative to a position of the rotor blades **140** during rotation about the rotor axis **550**.

[0050] FIG. **10** illustrates an example process **1000** for implementing flight control via the rotor hub of one or more embodiments described herein. The process may be used to adjust motor speed to achieve desired movement of a UAV, UAS, and/or other appropriate device or platform. The process may be performed whenever a UAV, UAS, and/or other appropriate device is launched or otherwise takes flight. In some embodiments, process **1000** may be performed by a device such as device **1100** described below, via an element such as rotor hub **100** or rotor hub system **500**.

[0051] As shown, process **1000** may include receiving (at **1010**) guidance information. Guidance information may be received in various appropriate ways, in various appropriate formats, and from various appropriate resources. For instance, in some cases, a UAV or other platform may have communication capabilities with a server, remote pilot or operator, and/or other appropriate devices or systems. Guidance information may include, for instance, direction and speed information relative to a current position of the UAV. As another example, guidance information may include global positioning system (GPS) coordinates, and/or other such navigation coordinates that may define a target or destination. As another example, guidance information may include data received via a user interface (UI), such as a joystick or similar control. The UAV or other associated platform may include various positioning sensors or components that may be able to provide guidance information (e.g., GPS sensors, accelerometers, gyroscopes, altimeters, etc.).

[0052] Process **1000** may include generating (at **1020**) a speed control signal. Based on the guidance information, the process may generate a speed control signal. Such a speed control signal may be similar to speed control signals **750**, **850**, or **950** described above. Generation of the speed control signal may include, for instance, determining a baseline or nominal rotational speed, as represented by speed control signal **750**. Such a nominal rotational speed may be based on various relevant factors, such as environmental factors (e.g., wind speed and/or direction, temperature, etc.), weight and/or payload associated with the UAV or other platform, and/or other relevant factors. Perturbations to the nominal rotational speed, such as pulses, sinusoidal signals, and/or other appropriate signals may be generated based on factors such as a current position, destination, directional controls (e.g., joystick signals), and/or other relevant guidance information. In some

embodiments, the speed control signal may be associated with a flight path, such that variations in direction, speed, and/or other attributes of movement may be automatically implemented as the UAV reaches each of a set of checkpoints. In some embodiments, a UAV or other platform may include sensors that may provide information relevant to speed control signal generation, such as environment sensors (e.g., temperature, humidity, etc.), force or stress sensors, etc.

[0053] The process may include determining (at **1030**) a rotor position relative to an axis of rotation (e.g., a position from zero to three-hundred-sixty-degrees). The rotor position may be determined in various appropriate ways. For instance, in some embodiments, the rotor position may be provided by a resource such as motor **510** (and/or be derived from signals received from motor **510**). In some embodiments, rotor hub **100** may include various position sensors and/or other similar elements that may be used to determine a rotor position.

[0054] As shown, process **1000** may include setting (at **1040**) rotor speed based on the rotor position and the speed control signal information. The rotor speed may be set by applying the generated speed control signal for the current rotor position. For example, the generated speed control signal may include a value for each increment of rotation (e.g., one-degree, one-half degree, one-tenth degree, etc.) and the value for the current rotor position may be applied via a resource such as a motor driver interface.

[0055] Process **1000** may include determining (at **1050**) whether updated guidance information is available. Such a determination may be made in various appropriate ways, depending on the type of guidance information available, sensors and/or other resources available, and/or other relevant factors. For instance, if the guidance information is provided as a destination having a set of GPS coordinates, an update message may be received from an external resource indicating a new destination with a different set of GPS coordinates, where such an update message may be identified as updated guidance information by the process. As another example, if the UAV associated with rotor hub **100** has moved more than a threshold distance since a last update, a new movement path may be calculated based on the destination location and current position. As another example, signals from a resource such as a joystick or other controller may be received at the UAV via a wireless communication link.

[0056] If process **1000** determines (at **1050**) that updated guidance information is not available, the process may repeat operations **1030-1050** until the process determines (at **1050**) that updated guidance information is available, or a target location is reached and/or other appropriate termination criteria are met.

[0057] If process **1000** determines (at **1050**) that updated guidance information is available, the process may include receiving (at **1060**) the updated guidance information. The updated guidance information may be received in a similar manner as the guidance information received at **1010**.

[0058] As shown, process **1000** may include generating (at **1070**) an updated speed control signal. The updated speed control signal may be generated based on the updated guidance information in a similar manner to the speed control signal generated at **1020**.

[0059] Process **1000** may repeat operations **1030-1070** until the process determines that a target location has been reached and/or other appropriate termination criteria are met.

[0060] One of ordinary skill in the art will recognize that process **1000** may be implemented in various different ways without departing from the scope of the disclosure. For instance, the elements may be implemented in a different order than shown. As another example, some embodiments may include additional elements or omit various listed elements. Elements or sets of elements may be performed iteratively and/or based on satisfaction of some performance criteria. Non-dependent elements may be performed in parallel. Elements or sets of elements may be performed continuously and/or at regular intervals.

[0061] The processes and modules described above may be at least partially implemented as software processes that may be specified as one or more sets of instructions recorded on a non-transitory storage medium. These instructions may be executed by one or more computational

element(s) (e.g., microprocessors, microcontrollers, digital signal processors (DSPs), application-specific integrated circuits (ASICs), field programmable gate arrays (FPGAs), other processors, etc.) that may be included in various appropriate devices in order to perform actions specified by the instructions.

[0062] As used herein, the terms “computer-readable medium” and “non-transitory storage medium” are entirely restricted to tangible, physical objects that store information in a form that is readable by electronic devices.

[0063] FIG. 11 illustrates a schematic block diagram of an exemplary device (or system or devices) **1100** used to implement some embodiments. For example, the systems, devices, components, and/or operations described above in reference to FIG. 1, FIG. 2, FIG. 3, FIG. 4, FIG. 5, FIG. 6, FIG. 7, FIG. 8, and FIG. 9 may be at least partially implemented using device **1100**. As another example, the process described in reference to FIG. 10 may be at least partially implemented using device **1100**.

[0064] Device **1100** may be implemented using various appropriate elements and/or sub-devices. For instance, device **1100** may be implemented using one or more personal computers (PCs), servers, mobile devices (e.g., smartphones), tablet devices, wearable devices, and/or any other appropriate devices. The various devices may work alone (e.g., device **1100** may be implemented as a single smartphone) or in conjunction (e.g., some components of the device **1100** may be provided by a mobile device while other components are provided by a server).

[0065] As shown, device **1100** may include at least one communication bus **1110**, one or more processors **1120**, memory **1130**, input components **1140**, output components **1150**, and one or more communication interfaces **1160**.

[0066] Bus **1110** may include various communication pathways that allow communication among the components of device **1100**. Processor **1120** may include a processor, microprocessor, microcontroller, DSP, logic circuitry, and/or other appropriate processing components that may be able to interpret and execute instructions and/or otherwise manipulate data. Memory **1130** may include dynamic and/or non-volatile memory structures and/or devices that may store data and/or instructions for use by other components of device **1100**. Such a memory device **1130** may include space within a single physical memory device or spread across multiple physical memory devices.

[0067] Input components **1140** may include elements that allow a user to communicate information to the computer system and/or manipulate various operations of the system. The input components may include keyboards, cursor control devices, audio input devices and/or video input devices, touchscreens, motion sensors, etc. Output components **1150** may include displays, touchscreens, audio elements such as speakers, indicators such as light-emitting diodes (LEDs), printers, haptic or other sensory elements, etc. Some or all of the input and/or output components may be wirelessly or optically connected to the device **1100**.

[0068] Device **1100** may include one or more communication interfaces **1160** that are able to connect to one or more networks **1170** or other communication pathways. For example, device **1100** may be coupled to a web server on the Internet such that a web browser executing on device **1100** may interact with the web server as a user interacts with an interface that operates in the web browser. Device **1100** may be able to access one or more remote storages **1180** and one or more external components **1190** through the communication interface **1160** and network **1170**. The communication interface(s) **1160** may include one or more application programming interfaces (APIs) that may allow the device **1100** to access remote systems and/or storages and also may allow remote systems and/or storages to access device **1100** (or elements thereof).

[0069] It should be recognized by one of ordinary skill in the art that any or all of the components of computer system **1100** may be used in conjunction with some embodiments. Moreover, one of ordinary skill in the art will appreciate that many other system configurations may also be used in conjunction with some embodiments or components of some embodiments.

[0070] In addition, while the examples shown may illustrate many individual modules as separate

elements, one of ordinary skill in the art would recognize that these modules may be combined into a single functional block or element. One of ordinary skill in the art would also recognize that a single module may be divided into multiple modules.

[0071] Device **1100** may perform various operations in response to processor **1120** executing software instructions stored in a computer-readable medium, such as memory **1130**. Such operations may include manipulations of the output components **1150** (e.g., display of information, haptic feedback, audio outputs, etc.), communication interface **1160** (e.g., establishing a communication channel with another device or component, sending and/or receiving sets of messages, etc.), and/or other components of device **1100**.

[0072] The software instructions may be read into memory **1130** from another computer-readable medium or from another device. The software instructions stored in memory **1130** may cause processor **1120** to perform processes described herein. Alternatively, hardwired circuitry and/or dedicated components (e.g., logic circuitry, ASICs, FPGAs, etc.) may be used in place of or in combination with software instructions to implement processes described herein. Thus, implementations described herein are not limited to any specific combination of hardware circuitry and software.

[0073] The actual software code or specialized control hardware used to implement an embodiment is not limiting of the embodiment. Thus, the operation and behavior of the embodiment has been described without reference to the specific software code, it being understood that software and control hardware may be implemented based on the description herein.

[0074] While certain connections or devices are shown, in practice additional, fewer, or different connections or devices may be used. Furthermore, while various devices and networks are shown separately, in practice the functionality of multiple devices may be provided by a single device or the functionality of one device may be provided by multiple devices. In addition, multiple instantiations of the illustrated networks may be included in a single network, or a particular network may include multiple networks. While some devices are shown as communicating with a network, some such devices may be incorporated, in whole or in part, as a part of the network.

[0075] Some implementations are described herein in conjunction with thresholds. To the extent that the term “greater than” (or similar terms) is used herein to describe a relationship of a value to a threshold, it is to be understood that the term “greater than or equal to” (or similar terms) could be similarly contemplated, even if not explicitly stated. Similarly, to the extent that the term “less than” (or similar terms) is used herein to describe a relationship of a value to a threshold, it is to be understood that the term “less than or equal to” (or similar terms) could be similarly contemplated, even if not explicitly stated. Further, the term “satisfying,” when used in relation to a threshold, may refer to “being greater than a threshold,” “being greater than or equal to a threshold,” “being less than a threshold,” “being less than or equal to a threshold,” or other similar terms, depending on the appropriate context.

[0076] No element, act, or instruction used in the present application should be construed as critical or essential unless explicitly described as such. An instance of the use of the term “and,” as used herein, does not necessarily preclude the interpretation that the phrase “and/or” was intended in that instance. Similarly, an instance of the use of the term “or,” as used herein, does not necessarily preclude the interpretation that the phrase “and/or” was intended in that instance. Also, as used herein, the article “a” is intended to include one or more items and may be used interchangeably with the phrase “one or more.” Where only one item is intended, the terms “one,” “single,” “only,” or similar language is used. Further, the phrase “based on” is intended to mean “based, at least in part, on” unless explicitly stated otherwise.

[0077] The foregoing relates to illustrative details of exemplary embodiments and modifications may be made without departing from the scope of the disclosure. Even though particular combinations of features are recited in the claims and/or disclosed in the specification, these combinations are not intended to limit the possible implementations of the disclosure. In fact, many

of these features may be combined in ways not specifically recited in the claims and/or disclosed in the specification. For instance, although each dependent claim listed below may directly depend on only one other claim, the disclosure of the possible implementations includes each dependent claim in combination with every other claim in the claim set.

Claims

1. A rotor hub comprising: a rotor hub hinge that rotates about an offset teeter axis; and a rotor coupling coupled to the rotor hub hinge, wherein the rotor coupling is able to couple to a rotor that rotates about an azimuthal rotational axis, and wherein the offset teeter axis is situated such that an offset angle between the azimuthal rotational axis and the offset teeter axis is between zero degrees and ninety degrees.
2. The rotor hub of claim 1, wherein the rotor hub is coupled to a pair of rotor blades.
3. The rotor hub of claim 1, wherein the offset angle is forty-five degrees.
4. The rotor hub of claim 1, wherein the rotor coupling comprises a frame with a horizontal member that is able to couple to a rotating portion of a motor and a vertical member that extends out from the horizontal member at a right angle.
5. The rotor hub of claim 4, wherein the rotor coupling further comprises a hinge coupling shaft that extends between the horizontal member and the vertical member at the offset angle relative to the horizontal member and the vertical member.
6. The rotor hub of claim 5, wherein the hinge coupling shaft couples to the rotor hub hinge via a cylindrical cavity running through the rotor hub hinge.
7. The rotor hub of claim 6, wherein an axis of rotation associated with the motor runs through the horizontal member.
8. An unmanned aerial system comprising: a rotor hub comprising: a rotor hub hinge that rotates about an offset teeter axis; and a rotor coupling coupled to the rotor hub hinge, wherein the rotor coupling is able to couple to a rotor that rotates about an azimuthal rotational axis, and wherein the offset teeter axis is situated such that an offset angle between the azimuthal rotational axis and the offset teeter axis is between zero degrees and ninety degrees; and a rotary motor comprising a stationary motor portion and a rotating motor portion, wherein the stationary motor portion is fixedly coupled to the unmanned aerial system and the rotating motor portion is coupled to the rotor coupling.
9. The unmanned aerial system of claim 8, wherein the rotor hub is coupled to a pair of rotor blades.
10. The unmanned aerial system of claim 8, wherein the offset angle is forty-five degrees.
11. The unmanned aerial system of claim 8, wherein the rotor coupling comprises a frame with a horizontal member that is able to couple to the rotating portion a vertical member that extends out from the horizontal member at a right angle.
12. The unmanned aerial system of claim 11, wherein the rotor coupling further comprises a hinge coupling shaft that extends between the horizontal member and the vertical member at the offset angle relative to the horizontal member and the vertical member.
13. The unmanned aerial system of claim 12, wherein the hinge coupling shaft couples to the rotor hub hinge via a cylindrical cavity running through the rotor hub hinge.
14. The non-unmanned aerial system of claim 13, wherein an axis of rotation associated with the motor runs through the horizontal member.
15. A method comprising: receiving guidance information; generating a speed control signal, wherein the speed control signal is associated with a rotor hub comprising: a rotor hub hinge that rotates about an offset teeter axis; and a rotor coupling coupled to the rotor hub hinge, wherein the rotor coupling is able to couple to a rotor that rotates about an azimuthal rotational axis, and wherein the offset teeter axis is situated such that an offset angle between the azimuthal rotational

axis and the offset teeter axis is between zero degrees and ninety degrees; and determining a rotor position; and setting a rotor speed based on the rotor position and the speed control signal.

16. The method of claim 15 further comprising: receiving updated guidance information; generating an updated speed control signal; determining the rotor position; and setting the rotor speed based on the rotor position and the updated speed control signal.

17. The method of claim 15, wherein the offset angle is forty-five degrees.

18. The method of claim 17, wherein the rotor coupling comprises a frame with a horizontal member that is able to couple to the rotating portion a vertical member that extends out from the horizontal member at a right angle.

19. The method of claim 18, wherein the rotor coupling further comprises a hinge coupling shaft that extends between the horizontal member and the vertical member at the offset angle relative to the horizontal member and the vertical member.

20. The method of claim 19, wherein the hinge coupling shaft couples to the rotor hub hinge via a cylindrical cavity running through the rotor hub hinge.
